18 DENSITY AND PRESSURE

One way of characterising materials is by their density. This chapter looks at density and at how materials can affect things around them by exerting pressure.





▲ Figure 18.1 Some effects of density and pressure

LEARNING OBJECTIVES

- Know and use the relationship between density, mass and volume:
  $\text{density} = \frac{\text{mass}}{\text{volume}}$ $\rho = \frac{m}{V}$
- Practical: investigate density using direct measurements of mass and volume
- Know and use the relationship between pressure, force and area:
  $\text{pressure} = \frac{\text{force}}{\text{area}}$ $p = \frac{F}{A}$
- Understand how the pressure at a point in a gas or liquid at rest acts equally in all directions
- Know and use the relationship for pressure difference:
  pressure difference = height × density × gravitational field strength
  $p = h \times \rho \times g$

UNITS

In this section you need to use degree Celsius (°C) and Kelvin (K) as the units of temperature (these units both represent the same change in temperature but the Kelvin scale starts from absolute zero, as explained later), kilogram per cubic metre (kg/m^3) as the unit of density, cubic metre (m^3) as the unit of volume, pascal (Pa) as the unit of pressure and square metre (m^2) as the unit of area. It is important to remember 1 m^3 = 1 000 000 cm^3 (a cubic metre is 1 million cubic centimetres) and 1 m^2 = 10 000 cm^2 (a square metre is 10 thousand square centimetres).

KEY POINT

ρ is the Greek letter rho (pronounced 'row') and is the usual symbol for density. This is not to be confused with p which represents pressure.

The properties of a material affect how it behaves, and how it affects other materials around it. The balloon can fly because the gas inside it has a very low density. The skis spread the weight of the skier over the snow so she does not sink into it. The submersible is designed to explore the seabed – it has a very strong hull to withstand the high pressure from water deep in the oceans.

DENSITY

Solids, liquids and gases have different properties and characteristics. One such characteristic is density. Solids are often very dense – that is, they have a high mass for a certain volume. Liquids are often less dense than solids, and gases have very low densities.

The density (ρ) of a material can be calculated if you know the mass (m) of a certain volume (V) of the material, using this equation:

$$\text{density, } \rho = \frac{\text{mass, } m}{\text{volume, } V}$$

$$\rho = \frac{m}{V}$$

The units for density depend on the units used for mass and volume. If mass is measured in kilograms and volume in cubic metres, the units for density are kilograms per cubic metre (kg/m^3). Density can also be measured in grams per cubic centimetre (g/cm^3).

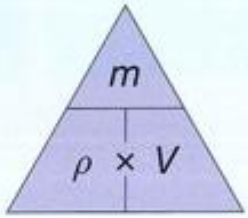
EXAMPLE 1

A piece of iron has a mass of 390 kg and a volume of 0.05 m^3. What is its density?

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$= \frac{390 \text{ kg}}{0.05 \text{ m}^3}$$

$$= 7800 \text{ kg/m}^3$$



▲ Figure 18.2 The equation for density can be rearranged using the triangle method.

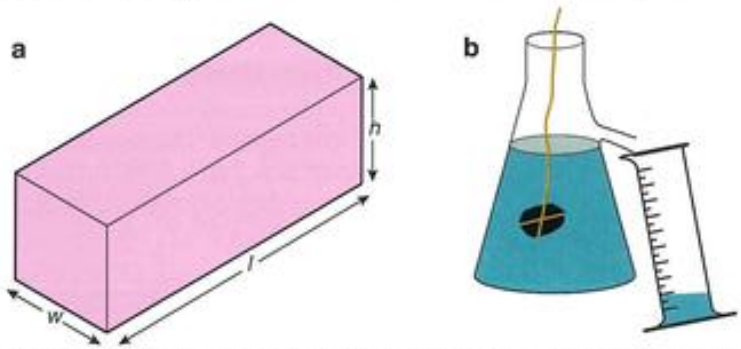
HINT

If an examination question asks you to write out the equation for calculating density, mass or volume, always give the actual equation in words or using standard symbols, such as $\rho = \frac{m}{V}$.

ACTIVITY

▼ PRACTICAL: INVESTIGATE THE DENSITY OF SOLIDS

The density of a substance can be determined by measuring the mass and volume of a sample of the material, and then calculating the density.



▲ Figure 18.3 **a** The volume of a regular solid can be calculated by multiplying its length (*l*), width (*w*) and height (*h*). **b** The volume of an irregular object can be determined using a displacement can and a measuring cylinder.

Use a half-metre rule to measure the length, width and height of a regular solid. When measuring the volume of the liquid that is displaced (pushed out) by an irregular solid make sure that the measuring cylinder is on a level (horizontal) surface and that you look at the scale straight on as shown (to avoid **parallax error**).

You then need to measure the mass of the solid. Use weighing scales for this.

EXAMPLE 2

The mass of 50 cm^3 of a liquid and a measuring cylinder is 146 g. The mass of the empty measuring cylinder is 100 g. What is the density of the liquid in kg/m^3?

mass of 50 cm^3 of liquid = 146 g – 100 g

= 46 g

= 0.046 kg

50 cm^3 = 0.000 05 m^3

$$\rho = \frac{m}{V}$$

$$= \frac{0.046 \text{ kg}}{0.000\,05 \text{ m}^3}$$

= 920 kg/m^3

or

$$= \frac{46 \text{ g}}{50 \text{ cm}^3}$$

= 0.92 g/cm^3

= 920 kg/m^3

HINT

Check the units when you are working out density – don't mix up g and m^3, for example. Also check which units the question asks for in your final answer.

1 kg = 1000 g

To convert kg to g, divide by 1000.

1 m^3 = 1 000 000 cm^3

To convert cm^3 to m^3, divide by 1 000 000 (or 10^6).

Alternatively, work out the density in g/cm^3, then multiply your answer by 1000.

PRESSURE UNDER A SOLID

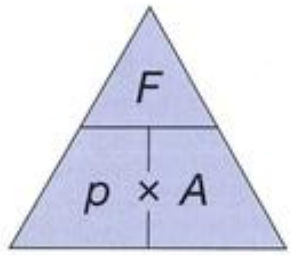
You can push a drawing pin into a piece of wood quite easily, but you cannot make a hole in the wood with your thumb, no matter how hard you push! The small point of the drawing pin concentrates all your pushing force into a tiny area, so the pin goes into the wood easily. Similarly, it is easier to cut things with a sharp knife than a blunt one, because with a sharp knife all the force is concentrated into a much smaller area.

Pressure is defined as the force per unit area. Force is measured in newtons (N) and area is measured in square metres (m^2). The units for pressure are pascals (Pa), where 1 Pa is equivalent to 1 N/m^2.

Pressure (p), force (F) and area (A) are linked by the following equation:

$$\text{pressure, } p \text{ (pascals)} = \frac{\text{force, } F \text{ (newtons)}}{\text{area, } A \text{ (square metres)}}$$

$$p = \frac{F}{A}$$



▲ Figure 18.4 The equation for pressure can be rearranged using the triangle method.

REMINDER

When answering questions, start with the equation you need to use in words or recognised symbols.

EXAMPLE 3

An elephant has a weight of 40 000 N, and her feet cover a total area of 0.1 m^2.

A woman weighs 600 N and the total area of her shoes in contact with the ground is 0.0015 m^2. Who exerts the greatest pressure on the ground?

Elephant:

$$p = \frac{F}{A}$$

$$= \frac{40\,000\ \text{N}}{0.1\ \text{m}^2}$$

$$= 400\,000\ \text{Pa (or 400 kPa)}$$

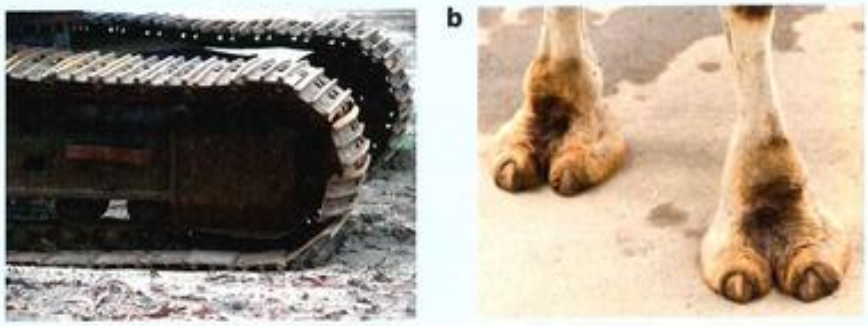
Woman:

$$p = \frac{F}{A}$$

$$= \frac{600\ \text{N}}{0.0015\ \text{m}^2}$$

$$= 400\,000\ \text{Pa (or 400 kPa)}$$

They both exert an equal pressure.



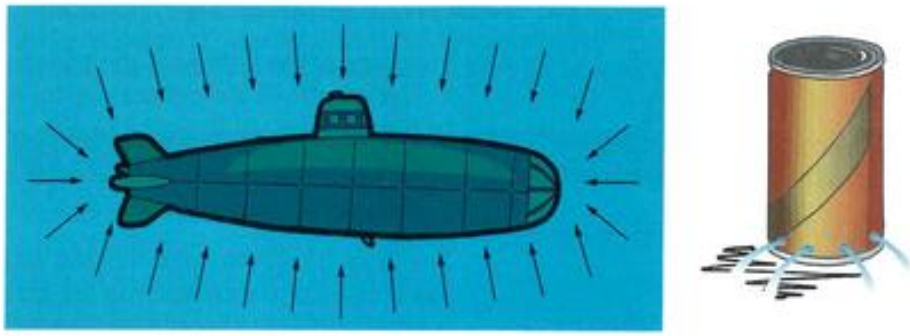
▲ Figure 18.5 **a** The caterpillar tracks on this vehicle spread its weight over a large area. **b** Camels have large feet so they are less likely to sink into loose sand.

Some machines, including cutting tools like scissors, bolt cutters and knives, need to exert a high pressure to work well. In other applications, a low pressure is important. Tractors and other vehicles designed to move over mud have large tyres that spread the vehicle's weight. The pressure under the tyres is relatively low, so the vehicle is less likely to sink into the mud. Caterpillar tracks used on bulldozers and other earth-moving equipment serve a similar purpose. (In Figure 18.5a the caterpillar tracks are very large.)

PRESSURE IN LIQUIDS AND GASES

The submersible shown in Figure 18.1 has a very strong hull to withstand the high pressure **exerted** on it by seawater. Pressure in liquids acts equally in all directions, as long as the liquid is not moving. You can easily demonstrate this using a can with holes punched around the bottom, as shown in Figure 18.6. When the can is filled with water, the water is forced out equally in all directions.

Gases also exert pressure on things around them. The pressure exerted by the atmosphere on your body is about 100 000 Pa (although the pressure varies slightly from day to day). However, the pressure inside our bodies is similar, so we do not notice the pressure of the air.



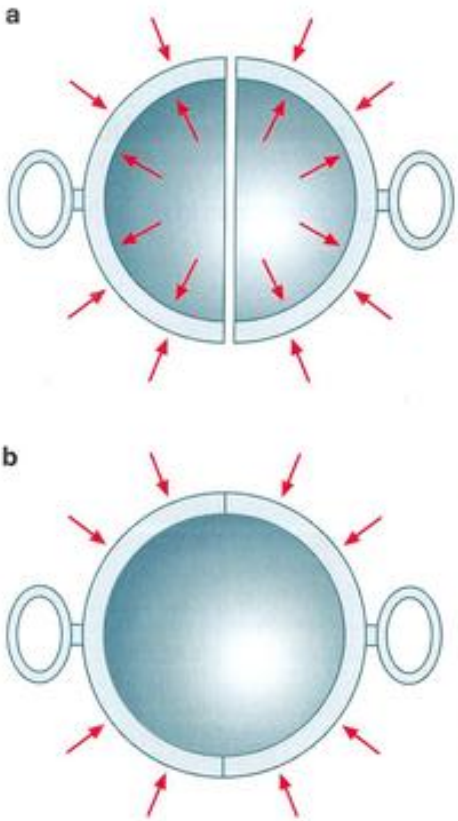
▲ Figure 18.6 Pressure in liquids acts equally in all directions. A can of water with holes can be used to demonstrate this.

One of the first demonstrations of the effects of air pressure was carried out by Otto van Guericke in 1654, in Magdeburg, Germany. Van Guericke had two large metal bowls made, put them together and then pumped the air out. The bowls could not be pulled apart, even when he attached two teams of horses to the bowls.



▲ Figure 18.7 Van Guericke's experiment at Magdeburg

You can do the same experiment in the laboratory, using much smaller bowls called Magdeburg hemispheres. When air is inside the spheres, the pressure is the same inside and outside. If the air is sucked out, pressure is only acting from the outside. The hemispheres cannot be pulled apart until air is let back into them.



▲ Figure 18.8 **a** When the hemispheres are full of air, the forces are the same inside and outside. **b** When the air is taken out, there is only a force on the outside of the hemispheres.

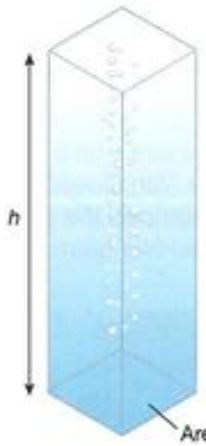
EXAMPLE 4

A laboratory set of Magdeburg hemispheres has a surface area of 0.045 m^2. What is the total force on the outside of the hemispheres?

$$F = p \times A$$
$$= 100\,000 \text{ Pa} \times 0.045 \text{ m}^2$$
$$= 4500 \text{ N}$$



▲ Figure 18.9 Pressure in a liquid increases with depth.



▲ Figure 18.10 We can work out the pressure difference between two points in a liquid by considering a column of the liquid.

KEY POINT

You need to learn this equation but you do not need to know the proof given here for the exam.

PRESSURE AND DEPTH

The experiment shown in Figure 18.9 demonstrates that the pressure in a liquid increases with depth. We can work out the pressure difference by thinking about a column of water, as shown in Figure 18.10.

The force at the bottom of the column is equal to all the weight of water above it. The volume of this water (V) is found by multiplying the area of its base (A) by the height (h) of the column. We can work out the mass (m) of the water by multiplying the volume ($A \times h$) by the density (ρ).

$$\text{mass of water, } m = (A \times h) \times \rho$$

The force (F) on the bottom of the water column is equal to the weight of this volume of water, which is the mass, m ($A \times h \times \rho$), multiplied by the gravitational field strength (g) (see page 34).

$$F = (A \times h \times \rho) \times g$$

As we are concerned with the pressure on the base of the column, we divide the force by the area:

$$F = \frac{A \times h \times \rho \times g}{A}$$

The area of the column therefore does not matter, and we can calculate the pressure difference between two points in a liquid using the equation:

pressure difference, p (Pa) = height, h (m) × density, ρ (kg/m^3) × gravitational field strength, g (N/kg)

$$p = h \times \rho \times g$$

Be careful not to muddle the symbol for pressure, p, with the symbol for density, ρ.

This equation can be used for calculating pressure differences in other liquids or gases, as long as you know their densities.

EXAMPLE 5

The first experiments on air pressure were carried out using a simple barometer. This was made by filling a long glass tube with mercury, and then inverting it into a bowl of mercury. The mercury falls until the weight of the column is supported by the pressure of the air on the mercury in the bowl.

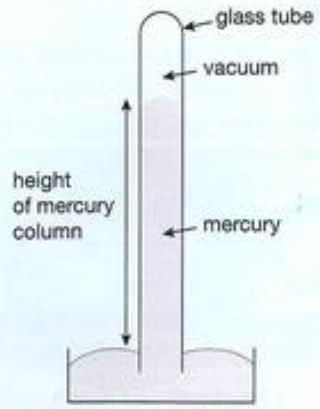
If the column of mercury is 0.74 m high, what is the air pressure?

The density of mercury is 13 600 kg/m^3. Take the gravitational field strength as 10 N/kg.

$p = h \times \rho \times g$

$= 0.74 \text{ m} \times 13\,600 \text{ kg/m}^3 \times 10 \text{ N/kg}$

$= 100\,640 \text{ Pa}$

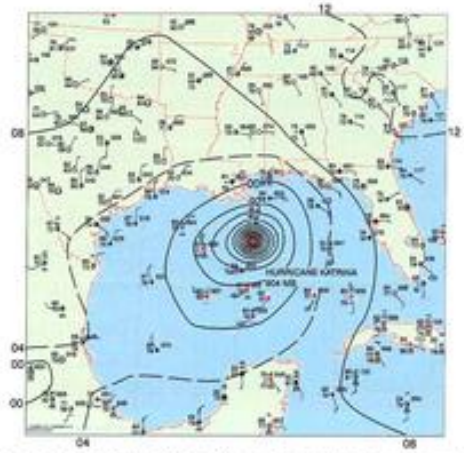


▲ Figure 18.11 A simple mercury barometer

EXTENSION WORK

The unit for pressure is named after Blaise Pascal (1623–1662). He was the first person to demonstrate that air pressure decreases with height. He persuaded his brother-in-law to take a mercury barometer up a mountain and measure the air pressure at different stages in the climb. He also arranged for someone at the base of the mountain to measure the air pressure at different times during the day. The measurements showed that the pressure of the air does decrease as you climb.

LOOKING AHEAD – WEATHER AND PRESSURE



▲ Figure 18.12 Weather maps show isobars (lines of equal air pressure). The centre of hurricane Katrina is a very low pressure region surrounded by isobars that are very close together. The result of the pressure difference is very strong winds.



▲ Figure 18.13 Tornados regularly cause huge amounts of damage in many parts of the world.

Variations in atmospheric pressure are partly responsible for the weather systems around the world. Meteorologists produce maps showing how atmospheric pressure varies from place to place to help them make predictions about future weather systems. Regions of low pressure produce winds as air flows from surrounding areas of high pressure.

Sometimes the wind speeds cause violent weather systems with tornadoes and hurricanes causing extensive damage and loss of life. Physics provides important tools to enable meteorologists to make more accurate predictions about when and where these devastating events will occur.

CHAPTER QUESTIONS

More questions on density and pressure can be found at the end of Unit 5 on page 193.

SKILLS CREATIVITY, REASONING

1 A Greek scientist called Archimedes was asked to check the purity of the gold in a crown. He did this by comparing the density of the crown with the density of pure gold.

(6)

a Describe how he could have measured the density of an irregular-shaped object such as a crown.

SKILLS PROBLEM SOLVING (8)

b Pure gold has a density of 19 000 kg/m^3. Suppose the crown had a volume of 0.0001 m^3. What mass should the crown be if it is made of pure gold?

2 People working on the roofs of buildings often lay a ladder or plank of wood on the roof. They walk on the ladder rather than the roof itself.

SKILLS REASONING (6)

a Explain why using a ladder or plank will help to prevent damage to the roof.

SKILLS PROBLEM SOLVING (7)

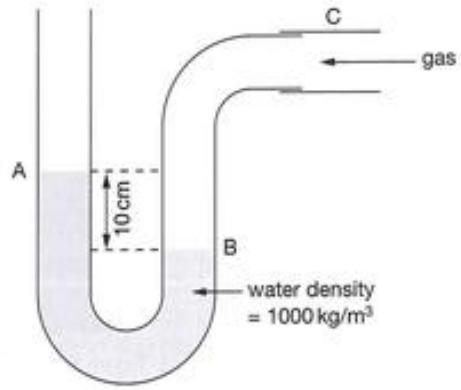
b A workman's weight is 850 N, and each of his boots has an area of 210 cm^2. Calculate the maximum pressure under his feet when he is walking. Give your answer in pascals.

c The workman lays a plank on the roof. The plank has an area of 0.3 m^2 and a weight of 70 N. What is the maximum pressure under the plank when he is walking on it?

HINT

Remember that all your weight is on one foot at some point while you are walking. To convert cm^2 to m^2, divide by 10 000 or 10^4.

3 A manometer can be used to find the pressure of a gas. The difference in the level of the liquid in each side of the tube indicates the difference in pressure on the water surface at A and the pressure of the gas on the water surface at B. (Assume atmospheric pressure is 100 kPa.)



a What is the difference in pressure in Pa between A and B? Use g = 10 N/kg.

b What is the pressure of the gas in tube C?

SKILLS REASONING

c Describe what will happen when the gas is turned off and the pipe is removed from C.